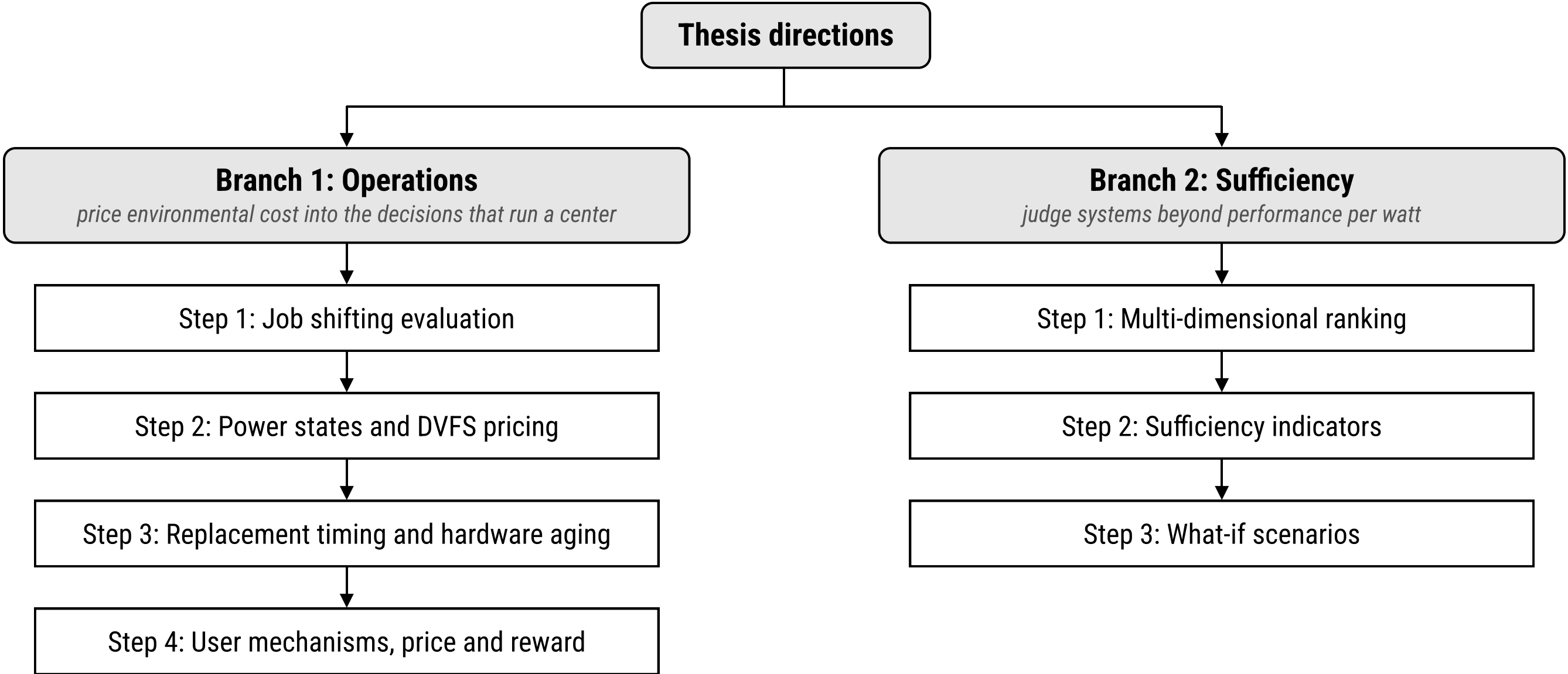


Thesis directions



```
graph TD; A[Thesis directions] --> B[Branch 1: Operations<br/>price environmental cost into the decisions that run a center]; A --> C[Branch 2: Sufficiency<br/>judge systems beyond performance per watt]; B --> D[Step 1: Job shifting evaluation]; D --> E[Step 2: Power states and DVFS pricing]; E --> F[Step 3: Replacement timing and hardware aging]; F --> G[Step 4: User mechanisms, price and reward]; C --> H[Step 1: Multi-dimensional ranking]; H --> I[Step 2: Sufficiency indicators]; I --> J[Step 3: What-if scenarios];
```

The diagram is a flowchart titled "Thesis directions". It branches into two main paths. The left path, "Branch 1: Operations", includes four steps: "Step 1: Job shifting evaluation", "Step 2: Power states and DVFS pricing", "Step 3: Replacement timing and hardware aging", and "Step 4: User mechanisms, price and reward". The right path, "Branch 2: Sufficiency", includes three steps: "Step 1: Multi-dimensional ranking", "Step 2: Sufficiency indicators", and "Step 3: What-if scenarios". Each branch box contains a subtitle: "price environmental cost into the decisions that run a center" for Branch 1 and "judge systems beyond performance per watt" for Branch 2.

Branch 1: Operations

price environmental cost into the decisions that run a center

Step 1: Job shifting evaluation

Step 2: Power states and DVFS pricing

Step 3: Replacement timing and hardware aging

Step 4: User mechanisms, price and reward

Branch 2: Sufficiency

judge systems beyond performance per watt

Step 1: Multi-dimensional ranking

Step 2: Sufficiency indicators

Step 3: What-if scenarios